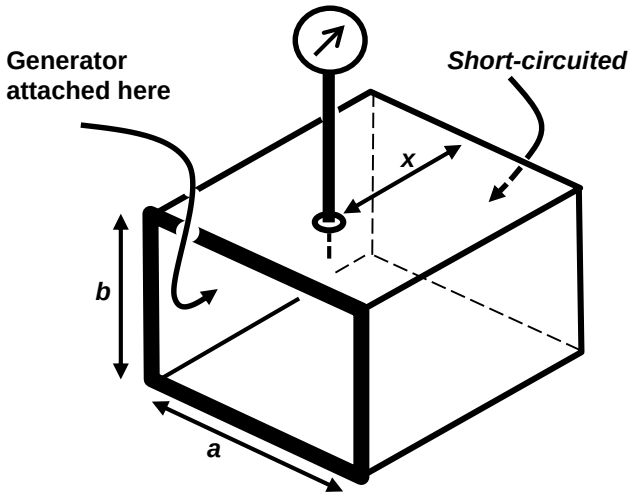
Winter term 2020/2021: Exercises "Radio Frequency Technology"

EXERCISE SESSION 6 (11.12.2020)

Problem 6.1: Guided wavelength

A rectangular waveguide with its inner dimensions a = 4 cm and b = 2.5 cm is given.

a) Calculate the cutoff frequency of the fundamental mode $TE_{10.}$
b) Calculate the upper limit of the monomode-frequency range, f_{max}.
c) The waveguide is now short-circuited at its end. The open side is connected to a generator with a matched waveguide. A probe which is able to measure the local electrical field strength is inserted into the waveguide through a hole. The hole is located in the middle of the side "***a***". The probe does not disturb the fields in the waveguide. (x = 5cm) Sketch the detected amplitude of the electrical field strength for the frequency range $0 < f < f_{max.}$ For which frequencies do the minimal and maximal field strength occur?
d) Now the waveguide is uniformly filled with another material ($\varepsilon_{rel} = \mu_{rel} = 2$). Recalculate the frequencies from problem part a), b) and c) with the new material.



Problem 6.2: waveguide loss

For a vacuum-filled rectangular waveguide of width a and height b, the conductor current loss is given as (script):

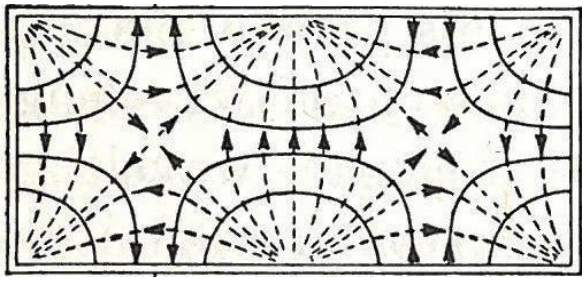
$$\alpha_{TE10} = R_{\square} \frac{1 + 2\,\frac{b}{a}\left(\frac{f_c}{f}\right)^2}{b\sqrt{\frac{\mu_0\,\mu_{rel}}{\varepsilon_0\,\varepsilon_{rel}}}\sqrt{1-\left(\frac{f_c}{f}\right)^2}} \quad , \qquad R_{\square} = \sqrt{\frac{\pi\, f\, \mu_0\, \mu_{rel}}{\sigma}} \quad [\Omega]\,, [\Omega/\square]$$

(a) For a waveguide with $a = 2b$, calculate the loss ratio at respective frequencies $2\,f_c$ and $1.1\,f_c$, where f_c is the cutoff frequency of the fundamental mode.
(b) For a waveguide WR28 with a = 7.112 mm, $a = 2\,b$, the measured loss is 1.45 dB/m at 30 GHz. Calculate the effective conductivity σ_{eff} of the material and compare it with the conductivity of brass, $\sigma = (0.23\ldots0.44) \times 58 \times 10^6$ S/m.

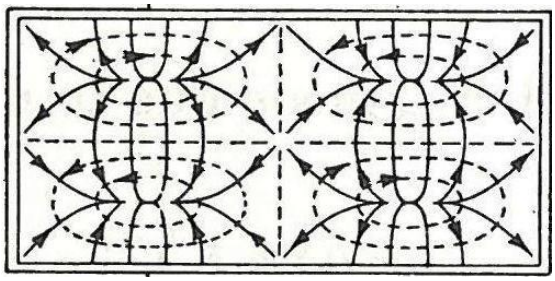
Winter term 2020/2021: Exercises "Radio Frequency Technology"

Problem 6.3: E and H field of waveguide modes

For the following two modes shown are the electric and magnetic field lines in the cross sectional plane, represented by solid and dashed lines, respectively) determine the mode type and designation (e.g., "TM_{94}") and the cutoff frequency for a = 22.86mm and b = 10.16mm.

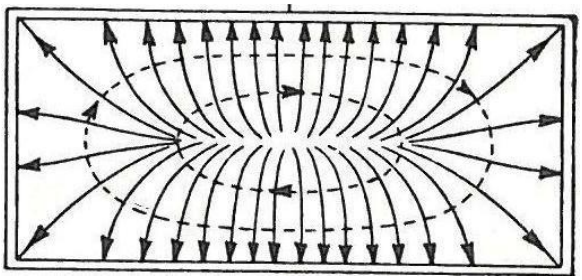


A1



A2

For the following mode (shown are the electric and magnetic field lines in the cross sectional plane, represented by solid and dashed lines, respectively) determine the mode type and designation (e.g., "TM_{94}") and draw the electric and magnetic field lines in the two longitudinal symmetry cut-planes.



Short questions session 6

a) A 100 MHz bandwidth signal is to be transmitted through a waveguide with a = 2b and cutoff frequency of the fundamental mode 3 GHz. The available bandwidths are:

a. 3.4-3.5 GHz
b. 5.4-5.5 GHz
c. 6.4-6.5 GHz

Which frequency band would you use? Why?

b) What is the relative bandwidth (with respect to fundamental mode) for the monomode range of a rectangular waveguide with a > 2b? And if a < 2b?

c) If you decrease the height of a rectangular waveguide, does the fundamental mode TE10 frequency cutoff change?

e) For the fundamental mode, does the longitudinal wavenumber β change? And what about transverse wavenumber k_T?

f) Is conductor loss per unit length higher, smaller or equal?

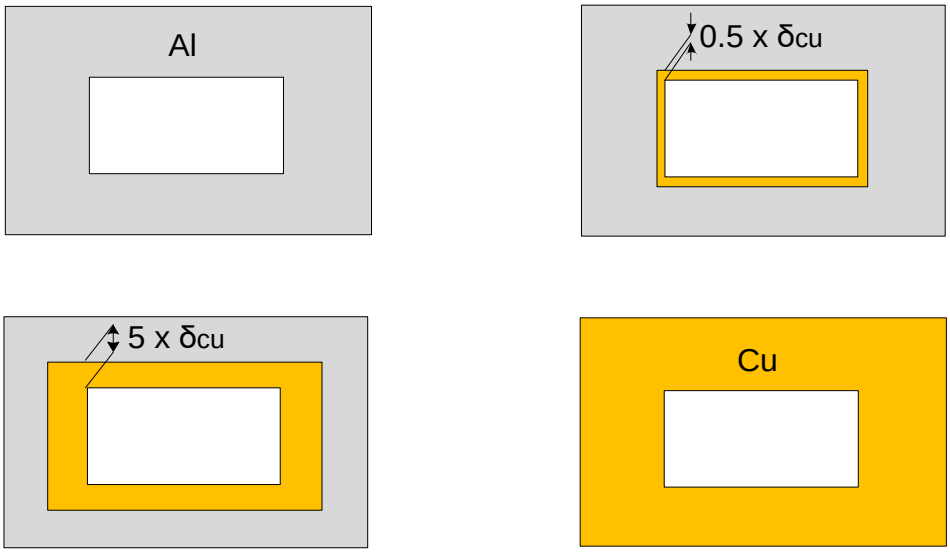
g) The skin depth δ measures how depth a wave is able to penetrate into a non-perfect conductor material (such as copper, gold, etc). It is measured as the length at which the amplitude of the field is decreased in 1/e.

Winter term 2020/2021: Exercises "Radio Frequency Technology"

h) How does skin depth change when frequency is increased?

i) What is the skin depth of a perfect electric conductor (PEC)?

j) The conductivity of copper is higher than aluminum's. Which material has higher skin depth? Which waveguide configuration would you choose from the next if the cost of aluminum is lower and keeping low conductor loss is important?



k) When probing a waveguide with a coaxial pin, should the pin be placed in an E-field maximum or rather in an H field maximum? For the answer given, should the pin stay parallel or perpendicular to the field?